ic_autolab round 7 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T04:23:34Z at git `62c2fb6bb27e715afe279b32539ea0a7494d193a`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-15a1d6` (`python3 tools/pcc/pcc.py standing PCC-20261104-15a1d6`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20	**regression**	ksum4-n8	1.026 [0.966, 1.090]	1.026 [0.962, 1.094]	1.025 [0.941, 1.115]	inc < sel	8.563 / 9.212 / 15.053	8.348 / 8.982 / 14.691	1.796 / 1.932 / 3.156	4.8x / 4.8x / 4.8x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

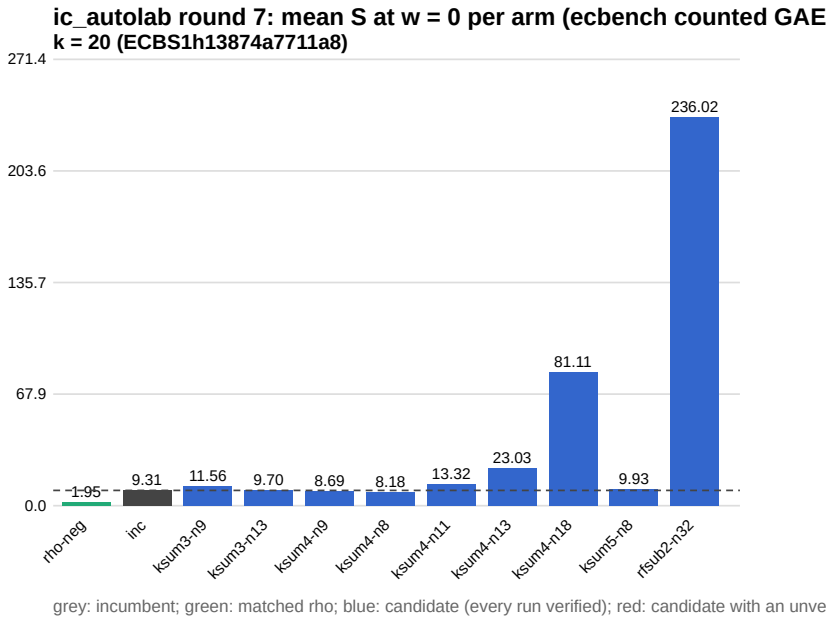
- k = 20: compare ops.status = ok, admissible = true; sel / inc = 1.0258 [0.966, 1.090] at w = 0; band: inc < sel; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 20: session `ECBS1h13874a7711a8` (run exit 0, verify exit 0, receipt sha256 `9fc538ecb45b3525b3525a1d7d625d3a07d1e953f0e2a515f573538f475d3f18`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	16	16	1.947 / 2.098 / 3.454	matched reference
`inc`	ksum3	11 `prime-projected-abscissa:size=11,scan_cap=176`	`ksum:m=3`	16	16	9.311 / 10.089 / 17.089	incumbent
`ksum3-n9`	ksum3	9 `prime-projected-abscissa:size=9,scan_cap=144`	`ksum:m=3`	16	16	11.562 / 12.817 / 24.115	incumbent neighbourhood: N = 0.85 x 11
`ksum3-n13`	ksum3	13 `prime-projected-abscissa:size=13,scan_cap=208`	`ksum:m=3`	16	16	9.703 / 10.336 / 16.028	incumbent neighbourhood: N = 1.2 x 11
`ksum4-n9`	ksum4	9 `prime-projected-abscissa:size=9,scan_cap=144`	`ksum:m=4`	16	16	8.690 / 9.117 / 12.961	cross-size transfer: ksum4 is incumbent at 3 sizes; ln N fitted on log2 r (slope 0.097) and extrapolated
`ksum4-n8`	ksum4	8 `prime-projected-abscissa:size=8,scan_cap=128`	`ksum:m=4`	16	16	8.184 / 8.774 / 14.079	cross-size transfer neighbourhood: N = 0.85 x 9 (ksum4 from k = 24)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`ksum4-n11`	ksum4	11	`prime-projected-abscissa:size=11,scan_cap=176`	`ksum:m=4`	16	16	13.319 / 13.605 / 16.170	cross-size transfer neighbourhood: N = 1.2 x 9 (ksum4 from k = 24)
`ksum4-n13`	ksum4	13	`prime-projected-abscissa:size=13,scan_cap=208`	`ksum:m=4`	16	16	23.028 / 23.275 / 25.491	cross-size transfer neighbourhood: N = 1.4 x 9 (ksum4 from k = 24)
`ksum4-n18`	ksum4	18	`prime-projected-abscissa:size=18,scan_cap=288`	`ksum:m=4`	16	16	81.114 / 81.393 / 83.910	cross-size transfer neighbourhood: N = 2 x 9 (ksum4 from k = 24)
`ksum5-n8`	ksum5	8	`prime-projected-abscissa:size=8,scan_cap=128`	`ksum:m=5`	16	16	9.930 / 10.408 / 14.713	exploration: family ksum5 untried at k = 20, N = 0.08 x N_a
`rfsub2-n32`	rfsub2	32	`prime-projected-abscissa:size=32,scan_cap=512`	`rootfind-subtract:m=2`	16	16	236.022 / 259.586 / 471.660	exploration: family rfsub2 untried at k = 20, N = 1 x N_a



train screen

Selection

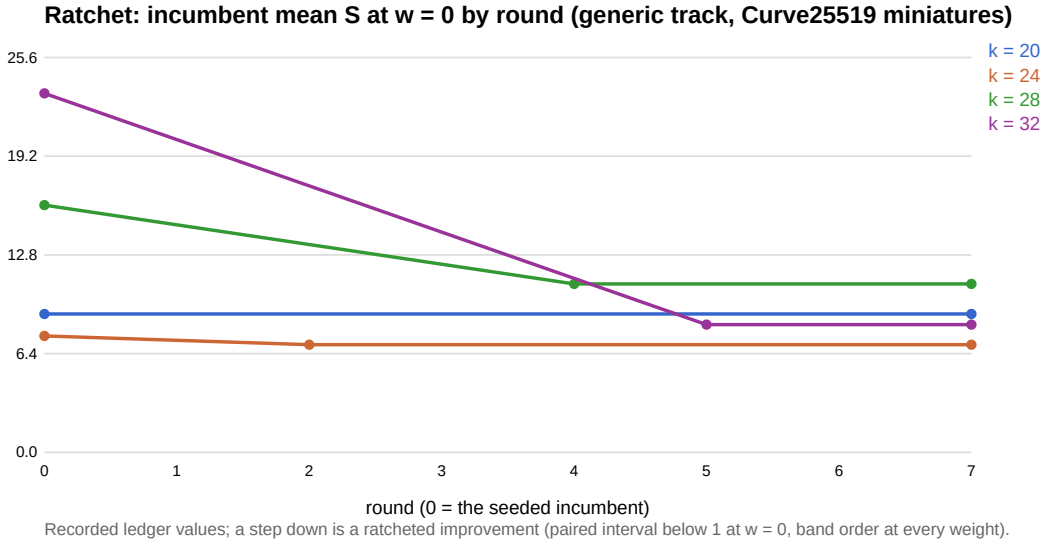
- k = 20: challenger `ksum4-n8` (Pareto alternative at w = 1: `ksum4-n9`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
20	ECBS1ha3ed7f4ce6e6	0	0	c5bc567be380c0c0e77d8e31f6b13c752dfeb0ec3d805ff6aca41959635cae1f	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	ksum3-n11 (prime-projected-abscissa:size=11,scan_cap=176 + ksum:m=3)	0	8.987 / 9.710 / 16.217	5.7x / 5.7x / 6.0x
24	ksum4-n12 (prime-projected-abscissa:size=12,scan_cap=192 + ksum:m=4)	2	6.985 / 7.514 / 12.271	6.0x / 5.9x / 5.6x
28	ksum4-n21 (prime-projected-abscissa:size=21,scan_cap=336 + ksum:m=4)	4	10.939 / 11.226 / 13.809	10.8x / 10.1x / 7.0x
32	ksum4-n26 (prime-projected-abscissa:size=26,scan_cap=416 + ksum:m=4)	5	8.296 / 8.811 / 13.446	9.1x / 8.8x / 7.4x



ratchet

Evidence and deviations

- ecbench binary sha256: `3fd22853485e388782ac0821295c56f81725fc832fd3610978be8088b5098bca`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.